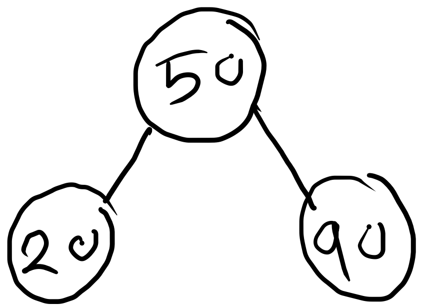
Red-black tree

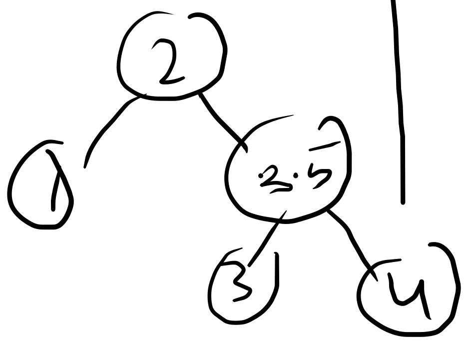
Home reading task

CLRS Book.

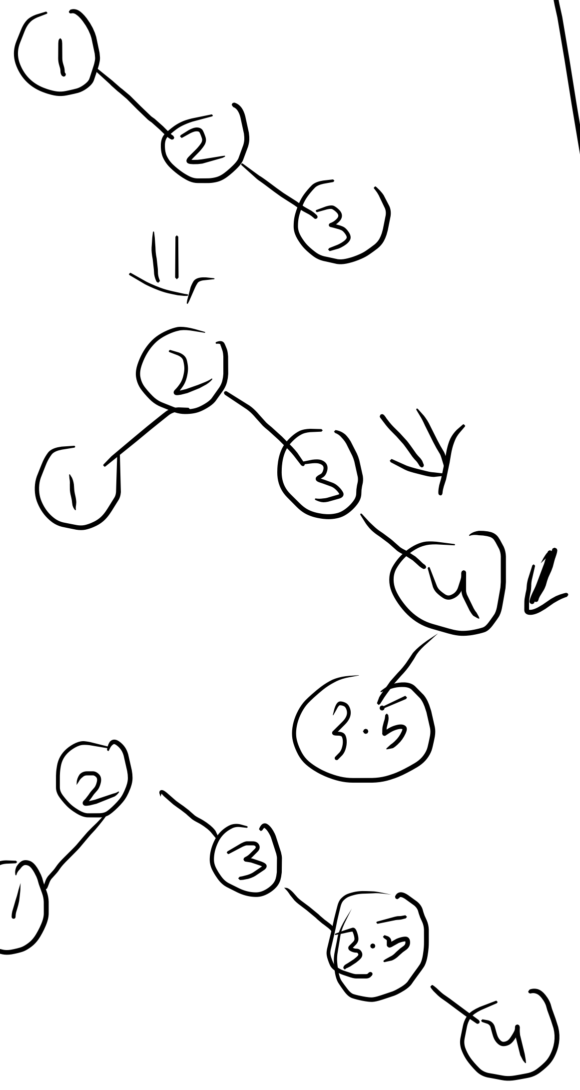
Example AVL tree

50 20 90 80

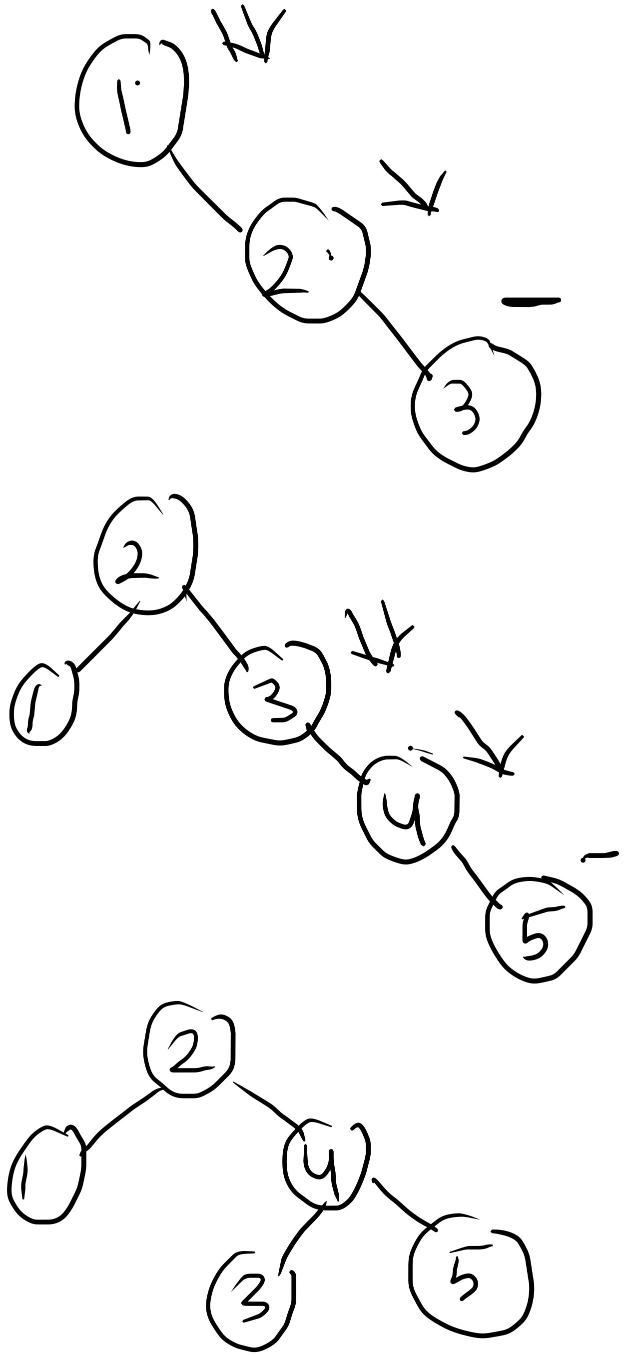




1 2 3 4 3.5



1 2 3 4 5



AVL Sorting - $O(n\log n)$

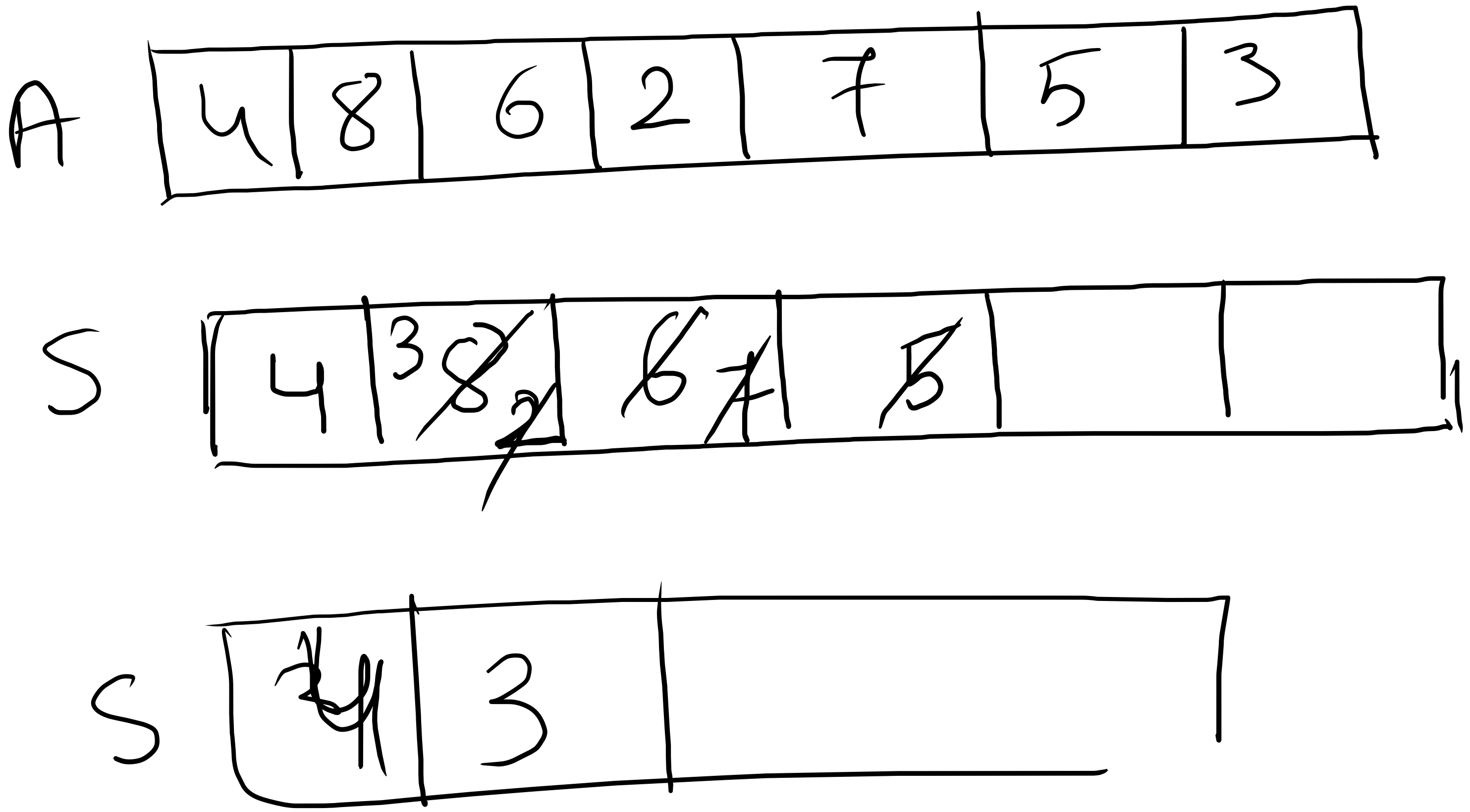
Dynamic array (Table doubling)

Multipop stack

```
Restricted-replication(A, S)
S is an empty stack
for i = 1 to length(A)            — n times.
    if A[i] ≤ size(S)             — θ(1)
        multi-pop(S, A[i])        — O(i)
    push(S, A[i])

multipop(S, K)
while notempty(S) and K > 0
    pop(S)
    K = K - 1

push(S, x) : standard push
pop(S)     : standard pop.
```



$$\sum_{i=1}^{n} O(i)$$

$$1 + 2 + \cdots + n = O(n^2)$$

Q: can it be $O(n)$??

Start with a empty stack.

and there are at most $\underline{n}$ operations:

push, pop and multi-pop.

$$n = n_{push} + n_{pop} + n_{multipop}$$

$$n_{pop} + n_{multipop} = O(n_{push})$$

$$\boxed{\text{number of pop element} \leq \text{number of push element or operation.}}$$

Total cost of n operations $\leq n_{push} + n_{push}$

$$= n_{push} \cdot O(1)$$

$$= O(n_{push})$$

$$= O(n)$$

$$\boxed{n_{push} \leq n}$$

Cost of a single operation $= \dfrac{O(n)}{n} = O(1)$

$\Rightarrow$ This type of analysis is called **amortised analysis**

Amortized analysis

→ Aggregate analysis   averaging

Total sum analysis.

- compute the worst case running time $T(n)$ of all the n operations in the sequence.

- Amortized cost per operation is $\frac{T(n)}{n}$

- All operations has the same amortized cost.

Binary counter increment

x 1 0 1 1 0 1 1 1

$\underline{\underline{x+1}}$ +   1

1 0 1 1 1 0 0 0

1 0 1 0 | 0 | 1 1 1 1

1 0 1 0 | 1 | 0 0 0 0